Sambhu Ganesan

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"I am only passionately curious" — Albert Einstein

Education

Lynbrook High School **Fall 2021 – present**
3.97 GPA

Math Course Work

Euler Circle: *2021 – Present*

Courses: Combinatorics, Analysis, Point-Set Topology, Number Theory, Analytic Number Theory

West Valley College: *2022 – Present*

Courses: Intermediate Calculus, Linear Algebra, Differential Equations

Programming experience

C, C++, Java, LaTeX.

Experience

Head Coach: Sam H. Lawson Middle School Mathcounts Club *2021 – Present*

Write handouts and lectures for students. Write and grade tests. Teach students math concepts weekly and have coached students to Mathcounts State's level from one of the hardest chapters in the country. Reference: Aletheia Chen.

Problem Writer and Test Coordinator: Mustang Math *2021 - Present*

Write problems for Mustang Math Tournament and shortlist worthy problem to appear on test. Contests have featured hundreds of contestants. Have test coordinated 2023 ASDAN, 2023 MMT, 2024 MMT and volunteered at 2024 MMT. Reference: Arpit Ranasaria.

Teaching Assistant: AlphaStar *June 2023 - July 2023*

Taught students competitive math at AIME level at one of the nations top performing learning centers(AlphaStar).

Co-Founder: PlantSTEM *2022 – Present*

Co-founder of 501(c) non-profit that teaches individualized classes for students at the entry level. Brought in top of the line instructors: USAPHO Quals, USAMO Quals, Nationally Ranked Chess players etc. Planted over 300 trees.

Researcher: Polymath Jr. *2024 – Present*

Accepted to an undergrad research math REU. Working on Skolem Sequences under Saad Mneimneh. Found novel results on Exact Skolem Sequences and Restricted Set Skolem Sequences. Invited as 1 of 6 participants to Joint Math Meeting (JMM), world's largest math conference, to present findings.

Researcher: Wolfram Summer Research Program *2024 – Present*

Accepted to Wolfram Summer Research Program as one of 64 students to do an independent project($< 10\%$ acceptance rate) . Did my project on Exploring Holomorphic Dynamics with Multiway Systems and found novel sets and patterns following a certain rule and proved them mathematically. Won 1st place at summer camp hackathon.

Researcher: Wolfram Emerging Leaders Program *2024 – Present*

Selected to Wolfram Emerging Leaders Program ($<5\%$ acceptance rate). Conducted a project on sky astronomy, focusing on pre-Galilean astronomy and the Moon's complex orbital mechanics, including synodic, sidereal, anomalistic, and draconic months. Explored the potential computational irreducibility of the Moon's orbit and analyzed patterns within these complexities. The project was led by Brian Silverman(MIT), a key developer of Scratch and Lego Mindstorms, and the president of the Playful Invention Company.

Awards

USAJMO: Qualifier
AIME: 4x Qualifier, Score: 9
AMC: Distinguished Honor Roll(top 1%), 4x Distinction(top 5%)
USAAAO Semifinalist / NAO: Qualifier
FBLA: Nationals

Projects and Writings

- Feynman Integration Technique
  S. Ganesan https://tinyurl.com/u66xn9wh

- LLL Algorithm
  S. Ganesan, N. Krishnan, I. Sun http://tinyurl.com/yc3emc7m

- Wiener-Ikehara Theorem
  S. Ganesan, I. Sun http://simonrs.com/eulercircle/analyticnt2024/sambhu-isaac-wiener-ikehara.pdf

- Exploring Holomorphic Dynamics With Multiway Systems
  S. Ganesan https://community.wolfram.com/groups/-/m/t/3215155

- CCD Astrometry of the Components of WDS 00153-2259
  S. Ganesan, B. Ancho, A. Nayak, E. Mun, P. Boyce, G. Boyce, M. Yokers http://www.jdso.org/volume20/number3/Ancho_377_387.pdf

- Harmful Brain Activity Classication of Spectrograms with Transfer Deep Learning
  S. Ganesan, S. Ram, Y. Kiran https://doi.org/10.21203/rs.3.rs-4294555/v1

- **Cage2Field**: Computer Vision *2023*
  Objective: Model baseball path onto the real field when batting in a batting cage. Used OpenCV to track and used kinematics and depth calculations to output parametric equations of the motion as a function of time. Then used linear algebra to transform 3-dimensional coordinates to 2D to graph on screen.

- **Anxy**: AI/ML *2024*
  Objective: Create a ML model with over 95% accuracy that analyzes text messages to classify depression/anxiety. If depression/anxiety is found then user will be connected to National Service Hotline. Used Wolfram Language.